An EV motor that estimates its internal heat when sensors fail

Detailed project proposal | 24 October 2026

Academic title: Fault-Tolerant Thermal State Estimation for Electric Vehicle Traction Motors Using Physics-Guided Deep Learning.

Working name: ThermoGuard Motor. **Domain:** electrical drives, motor thermal modeling, deep learning. **Initial platform:** a permanent-magnet synchronous motor (PMSM) test-bench dataset.

Project in one paragraph

The proposed system will estimate internal motor temperatures from electrical operating signals and accessible boundary temperatures. Its main purpose is to maintain useful temperature estimates when an input sensor stops reporting. A compact neural thermal model will combine a heat-transfer structure with learned nonlinear relationships. The project will compare ordinary training with training that deliberately masks sensor measurements, then assess conservative temperature bounds on entirely unseen measurement sessions. A dashboard will replay operating profiles and display the estimated rotor-magnet and stator temperatures, sensor availability, and an advisory torque limit. Recorded measurements will provide evaluation references. A separate optional simulator will assess the consequences of changing torque. The project will produce reproducible experiments and an evidence-based paper, including negative findings if the proposed method does not improve the baselines.

Central research question

Can training a physics-guided thermal estimator to handle missing inputs reduce dangerous temperature underestimation during sensor dropouts on unseen motor operating profiles, while preserving useful accuracy during normal operation?

Proposed outputs

- A documented data pipeline with splits by complete measurement session.
- Conventional, recurrent-neural, and thermal-neural reference models.
- Reproducible sensor-dropout experiments and an uncertainty evaluation.
- A replay dashboard with measured references available only for evaluation.
- A detailed paper and presentation supported by the measured results.

Scope of this document

This is a proposal and implementation plan. The dataset archive has not yet been downloaded or audited locally. No model has been trained and no accuracy, protection benefit, or deployment performance has been measured. The literature establishes that neural motor-temperature estimation already exists; the proposed investigation concerns reliability during missing measurements [R1-R5].

1. Electrical engineering basis and scope

Why the problem matters

Internal temperatures reflect accumulated losses and cooling history. A motor can remain hot after its instantaneous electrical load falls. Estimating temperatures therefore requires a dynamic state, rather than a regression using only the latest current and speed. The physical motivation is established in the motor thermal-modeling literature [R2, R3].

The project will estimate four instrumented evaluation targets: rotor permanent magnet (`pm`), stator winding, stator tooth, and stator yoke. These are component-level estimates, not a spatially resolved finite-element temperature field.

Physical model

For thermal node k, a lumped heat balance is:

```
C_k dT_k/dt = P_loss,k + sum_j G_kj (T_j - T_k)
```

Here C is thermal capacitance, G is thermal conductance, and P_loss is generated heat. The proposed model will constrain capacitances and conductances to be positive and loss estimates to be nonnegative. These constraints improve physical plausibility but do not establish accurate material parameters or certified protection.

Useful electrical features include `i_d^2 + i_q^2`, `u_d*i_d + u_q*i_q`, and speed magnitude. The second quantity is an electrical-power proxy whose scaling depends on the d-q transformation convention. It must not be reported as measured motor efficiency. Regeneration still generates losses, so negative electrical power must not become negative heat generation.

Primary operating inputs

Signal group	Inputs	Reason for inclusion
Electrical	i_d, i_q, u_d, u_q	Load and electrical-excitation information
Mechanical	motor_speed	Operating point and speed-dependent behavior
Cooling boundary	coolant	Accessible boundary-temperature measurement
Environment	ambient	Ambient-temperature boundary condition

Measured shaft torque will be excluded from the primary estimator because it may require an instrument absent from a production drive. It may support descriptive analysis and an explicitly labeled upper-bound experiment. The four internal target temperatures will never be ordinary model inputs.

Scope boundary

The core project covers one measured motor platform, missing-data robustness, and temperature estimation. Cross-vehicle transfer, mechanical fault classification, winding hotspot reconstruction, certified protection, and full-vehicle range improvement are outside the initial scope. The 2 Hz data will support thermal-timescale analysis, not switching-frequency fault detection [R1].

2. Research contribution and data protocol

What is established and what will be investigated

Established foundation	Proposed investigation
Deep recurrent and convolutional motor-temperature estimators [R2]	Behavior when an operating-input sensor becomes unavailable
Thermal neural networks with a heat-transfer structure [R3]	Effect of explicit missing-input masks and measurement-age features
Hybrid thermal/deep models and sparse-sampling studies [R4, R5]	Underestimation, uncertainty, and clean-versus-fault performance under one consistent protocol

The project does not claim to invent thermal neural networks. Its candidate contribution is a reproducible reliability study and, if supported by experiments, an improved strategy for handling missing inputs. A focused review must check this exact combination before the final paper claims novelty.

Dataset and provenance

The primary source is the Paderborn Electric Motor Temperature dataset. Its public card describes approximately 185 hours of test-bench recordings at 2 Hz, separated by `profile_id` [R1]. At acquisition, record the downloaded version, column schema, units, checksum, license, and source URL. Do not redistribute the archive until its actual license terms have been verified.

The source contains measured target temperatures for evaluating a virtual sensor. Artificial sensor faults will be introduced into copies of the input channels. Those faults are controlled experimental perturbations, not evidence that the original motor suffered physical faults.

Separation before preprocessing

Use approximately 50% of whole profiles for training, 10% for validation, 20% for calibration, and 20% for final testing. Round by available profile count and publish the exact IDs in a manifest. Calibration should contain at least nine independent profiles for the proposed finite one-sided 90% profile-level bound. If too few profiles remain, revise the split or report empirical intervals instead.

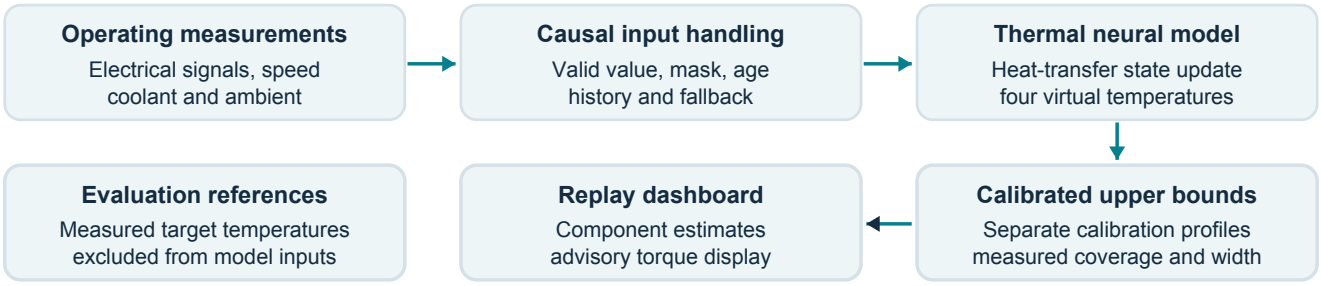
Fit normalization, imputation constants, and feature transformations on training profiles only. Select architecture and early stopping using validation profiles. Use calibration profiles only to set the uncertainty correction. Open the test set after the experiment settings are frozen.

Causal processing and initial conditions

Preserve row order within each profile. Do not create a time window that crosses a profile boundary, interpolate using future samples, or split overlapping windows across datasets. Reset model state at every new profile.

Initial internal temperatures are not observable from a single input sample. Start from an explicitly declared boundary-based estimate and report initial-transient errors separately, as well as full-profile errors. A study using measured initial target temperatures may appear only as an instrumented upper bound. Retaining a previous estimate is valid only within the same continuous session. Report a first-five-minute sensitivity analysis; this duration is a proposed diagnostic window, not a claim about physical settling time.

3. Model and sensor-failure design



Reference models

B0: simple dynamic regression. Fit a regularized linear estimator using causal exponentially weighted electrical features. This establishes how much benefit comes from thermal history without a deep architecture.

B1: compact GRU. Use the same operating inputs and target temperatures, with no target-temperature feedback during deployment. It provides a conventional temporal deep-learning comparison.

B2: thermal neural network. Reproduce a compact version of the authors' public heat-transfer-based architecture and document all deviations [R3].

Proposed B3: thermal neural network with missing-input training. Add a validity mask and elapsed time since the last valid value for each input. Train on clean sequences mixed with declared dropout patterns. Compare B3 directly with B2 using the same backbone and optimization budget.

Core failure protocol

Scenario	Proposed perturbation	Interpretation
F0: clean	Original operating signals	Accuracy cost of robustness
F1: coolant dropout	Remove one channel for 5, 30, or 120 s	Missing boundary-temperature measurement
F2: voltage dropout	Remove u_d or u_q for the same durations	Missing electrical telemetry
F3: combined dropout	Remove coolant and one voltage channel	Harder sensor combination
F4: unreported errors	Coolant bias ramp or stuck value	Stress test without an oracle fault flag

The durations are proposed experimental settings, not dataset facts. Fault start times and seeds must be independent of target temperature. Keep unmodified reference channels and target values in evaluation-only storage.

What the estimator knows

For F1-F3, use missing-sample flags that could come from the acquisition system. Fill inputs causally with a short last-valid-value hold, followed by a training-derived fallback; always expose the mask and age. Learn hold durations on validation data and freeze them before testing.

F4 is different: the sensor still reports a value, so do not provide the model with the injected onset or true fault mask. Measure degradation under bias and stuck-value stress. Automatic drift diagnosis is an extension requiring a separately evaluated detector. Naturally steady values must not automatically count as failed sensors.

Training and numerical behavior

Use a temperature-regression loss, initially Huber, with a validation-selected penalty on positive underestimation `max(T_true - T_est, 0)`. Keep the penalty in a separate ablation to avoid attributing all gains to masking. Use a stable thermal integration step and check long-rollout behavior. Train all state transitions causally; do not substitute measured targets as hidden states during final evaluation.

4. Evaluation and uncertainty

A fair comparison

Evaluate all models on identical complete test profiles and identical injected events. Train three random seeds per model when resources permit. Report variability across training seeds separately from uncertainty across test profiles. A final model selection must use validation data, never the best test result.

Metric	Engineering interpretation
MAE and RMSE, in degrees Celsius	Average temperature-estimation error
95th percentile positive underestimation	Tail of errors that hide overheating
Maximum positive underestimation	Worst observed miss, with its profile and time
Upper-bound coverage and width	Conservatism versus usefulness
Recovery after measurement return	Estimator behavior after a dropout
CPU inference time and model size	Feasibility on the tested computer

Compute each metric per target and per profile. Report macro averages across profiles, pooled metrics, and the worst profile. Avoid reporting only R-squared or a percentage called temperature-prediction accuracy.

Proposed conservative upper temperature bound

Train a small ensemble of three B3 models. Let `mu` be the ensemble mean and `sigma` its standard deviation. Use a small positive scale floor selected on validation data. For each calibration profile, calculate the maximum positive standardized error over all targets, all times, and a fixed suite of clean/dropout replays:

```
s_g = max(0, max_(t,k,scenario) ((T_true - mu) / max(sigma, floor)))
```

For alpha = 0.10, take the order statistic with index `ceil((n_cal + 1)*(1-alpha))`. If the required index exceeds the number of calibration profiles, a finite distribution-free bound is unavailable. Otherwise define `U = mu + q*max(sigma, floor)`.

This profile-level construction accommodates dependence within each profile by treating a complete profile as one calibration unit. Its nominal interpretation requires exchangeable calibration and test profiles under the same replay protocol. New motors, unreported drift, or a changed fault distribution invalidate that assumption. Report their empirical coverage separately. Ensemble disagreement alone is not a calibrated probability.

Success criterion and reporting

The primary comparison is B3 versus B2 on the predeclared F1-F3 suite. The candidate target is at least a 15% reduction in macro-averaged 95th-percentile positive underestimation, with no more than a 10% increase in clean macro MAE. These are proposed engineering targets, not expected or achieved results. Also report the absolute errors and profile-bootstrap intervals; percentage gains near a zero baseline can mislead.

A useful negative result is acceptable: if missing-input training fails to improve the baseline, explain which signals or initial conditions make the state insufficiently observable. The research contribution then becomes a transparent benchmark and its limitations.

5. Demonstration, control extension, and research limits

Dashboard specification

The initial application will replay measured profiles. Its main panel will display a schematic of the rotor magnet, winding, tooth, and yoke, colored by estimated component temperature. Call it a component-temperature schematic, not a measured spatial heatmap. A trace panel will show estimates and upper bounds, with ground truth available through an evaluation toggle. A sensor panel will show channel availability and time since the latest valid reading.

The operator will choose a held-out profile, inject a declared dropout, and compare the clean-trained and fault-trained estimators. A visible label will distinguish injected data faults from physical motor faults. Each replay will record its profile, seed, model checkpoint, and settings.

Advisory torque limit

An optional display can translate each upper temperature bound into a torque multiplier:

```
r_k = clip((L_k - U_k)/(L_k - W_k), 0, 1)

r = min_k r_k
```

W and L are component-specific warning and stop settings, with W below L. Production values require motor specifications and engineering validation. Until those exist, any selected settings are illustrative and must appear as such. The dataset's torque channel is measured shaft torque, not a requested torque command.

What recorded replay can demonstrate

Replay can show when an advisory would have reduced a hypothetical request. It cannot show that torque reduction actually cooled the motor, prevented a violation, or preserved vehicle performance, because the recorded temperatures do not respond to the new command.

For a control-effect study, implement a separate motor/thermal plant with a synthetic requested-torque profile. Perturb plant parameters independently of the estimator and evaluate thermal-limit exceedance, delivered mechanical work, and unnecessary derating. Report all results as simulation-dependent. Retain independent hard limits in any future hardware experiment.

Limitations to carry into the final paper

- One motor platform does not demonstrate cross-motor or cross-manufacturer transfer.
- Injected signal faults do not reproduce all physical sensor or cooling failures.
- Sparse inputs cannot uniquely reveal every unknown initial thermal condition.
- Low average error can coexist with a dangerous isolated underestimation.
- Nominal uncertainty coverage does not establish coverage under arbitrary distribution shifts.
- A software timing measurement does not establish execution time on an automotive controller.

The paper will use the terms "virtual thermal sensor" and "thermal estimator." A live digital-twin or real-time deployment claim requires the corresponding integration and timing evidence.

Implementation plan

Planning basis: approximately 12 academic weeks, with a working estimation/replay prototype targeted by week 8. The schedule is a proposed effort estimate, not a delivery commitment. The same person can combine roles; a three-person team can divide electrical modeling, machine learning, and software/evaluation.

1. Planned software organization

The following structure describes future implementation. The modules and commands are not yet implemented.

Planned path	Responsibility
data/raw/	Locally acquired, unchanged source archive
data/manifests/	Checksums, schema, license record, profile splits
src/data.py	Causal features, normalization, profile-safe sequences
src/faults.py	Deterministic dropout, bias, and stuck-value events
src/models/	Dynamic regression, GRU, and thermal neural models
src/train.py	Seeded training, checkpoints, validation selection
src/calibrate.py	Profile-level ensemble-bound calibration
src/evaluate.py	Clean/fault metrics and profile-bootstrap intervals
src/supervisor.py	Illustrative advisory torque multiplier
src/plant.py	Optional independent thermal/control simulator
app.py	Recorded-data replay dashboard
configs/	Frozen model, split, and failure-suite settings
tests/	Leakage, causality, fault, and calibration checks
outputs/	Results, figures, manifests, and experiment logs

Recommended tools

Use Python, PyTorch, NumPy, pandas, scikit-learn, matplotlib, and Streamlit. Add pytest for meaningful data and state-transition tests. Pin versions after a successful installation and smoke test on the actual host. These packages are recommendations; this proposal does not assume they are all installed.

A modern laptop with approximately 16 GB RAM is a reasonable starting point. A GPU is helpful for the ensemble but should be optional for small models. Begin with one target and a few training profiles, then expand to four targets and full profiles. Measure resource consumption before increasing model size. No EV, traction inverter, or high-voltage hardware is required for the core recorded-data study.

Run records

Every experiment should retain the source checksum, git revision, exact profile split, preprocessing statistics, fault schedule, random seed, model settings, checkpoint selection rule, and output metrics. Keep source data out of version control unless redistribution is explicitly permitted. Use the same event manifest for every model in a paired comparison.

2. Work packages and weekly milestones

Week	Work package	Reviewable completion evidence
1	Acquire and audit data; verify license and literature	Dataset card, units, checksum, profile inventory, source matrix
2	Freeze grouped splits; implement causal preprocessing	Split manifest and passing leakage/causality checks
3	Train dynamic regression and a small GRU	Clean per-profile metrics and reproducible checkpoints
4	Implement and validate thermal neural baseline	Documented thermal update, stable full-profile rollouts
5	Add missing-input masks and causal fallback	Deterministic F1-F3 events and B2/B3 comparison
6	Train ensemble and calibrate upper bounds	Calibration record, coverage and interval-width plots
7	Run frozen test suite and bias/stuck stress tests	Complete metrics, seeds, uncertainty, worst-case traces
8	Build replay dashboard	A full demonstration with logged settings and exportable figures
9	Optional independent plant and derating study	Clearly labeled simulated control-effect results
10	Repeat only unresolved experiments; interpret outcomes	Locked result tables, ablations, limitations
11	Write research paper using measured results	Manuscript, bibliography, methods and data statement
12	Prepare presentation and reproduce a clean run	Slide deck, demonstration script, handover instructions

Stage gates

Gate A, end of week 2: data is accessible, units are understood, and profile isolation is proven. If access fails, switch to the authors' documented acquisition route or a separately licensed source; do not silently replace measured data with synthetic samples.

Gate B, end of week 4: clean temperature estimates are reproducible. If the thermal model is unstable or too costly, use the GRU as the primary estimator and retain physics-guided work as an extension. Explain the scope change in the paper.

Gate C, end of week 7: test metrics and known failure cases are available. If fault robustness does not improve, retain the benchmark and investigate observability, rather than inventing gains or tuning to the final test set.

Gate D, end of week 8: a reviewer can replay an unseen profile, trigger a dropout, and inspect estimates without a live model-training session.

Gate E, week 9: add the control simulator only if the core estimation study is complete. A polished estimation project is sufficient; the optional control study must not displace the main evidence.

3. Experiment matrix and implementation details

Minimum model comparison

Model	Missing-input exposure	Main purpose
B0: dynamic linear regression	Same causal fallback at test time	Simple temporal baseline
B1: GRU	Clean training initially	Standard deep sequence baseline
B2: thermal neural network	Clean training	Physical-structure baseline
B3: same network plus masks and age	Clean and masked training	Isolate fault-training benefit
B4: B3 with underestimation penalty	Same fault training	Isolate asymmetric-loss effect

The ensemble is a calibration layer over the selected robust model, not a replacement for the single-model comparisons. Select B3 or B4 on validation results before final testing. Keep B2/B3 backbones identical except for the features needed to represent missingness. Include a mask-without-fault-training ablation if the apparent benefit could come from added parameters.

Event generation

Pre-register event durations of 5, 30, and 120 seconds. At 2 Hz these correspond to 10, 60, and 240 samples if the acquisition audit confirms uninterrupted 0.5-second sampling. Place events away from profile boundaries and record the duration actually available. Train on selected durations and combinations; reserve one harder duration/combination for a declared generalization stress test.

For stuck values, replace future observations with the last valid sample at onset. For a bias ramp, add a declared ramp to the chosen input channel. Suggested coolant stress magnitudes are +5 and -5 degrees Celsius over 120 seconds, with a separate +10/-10-degree severity test. These values are experimental assumptions. Do not corrupt target labels or supply an oracle mask for these unreported errors.

Recovery and cold-start evaluation

When a channel returns, record the temperature-error trajectory and compare it with the same model's clean replay. Report a recovery curve; define any recovery-time threshold using validation data before testing. Evaluate cold starts with boundary-based initialization. Report an instrumented initial-state experiment separately if performed.

Statistical comparisons

Calculate paired model differences using the same profiles and event schedules. Bootstrap complete test profiles, not individual timestamps, to obtain intervals for aggregate differences. Keep training-seed variation separate. If a session contains multiple related profiles or repeats, group by the broader session identifier when available. Operating-profile holdout on one motor supports same-platform generalization only.

4. Quality checks, risks, and completion criteria

Meaningful implementation checks

- Assert that training, validation, calibration, and test profile IDs are disjoint.
- Confirm that target-temperature columns and measured torque cannot enter primary input features.
- Verify that altering a future sample cannot change earlier model predictions.
- Confirm that window extraction never spans two profiles and recurrent state resets at boundaries.
- Ensure generated dropout events leave the reference labels unchanged and are repeatable from a seed.
- Check that an unreported bias/stuck event does not expose its true onset or mask to the estimator.
- Test heat-flow sign conventions, positive thermal parameters, and stable long rollouts.
- Confirm that upper-bound calibration uses calibration profiles only and handles insufficient sample counts explicitly.
- Verify that replay remains advisory and does not report counterfactual cooling as an observed outcome.

Risk register

Risk	Response
Dataset access or unclear license	Resolve access and terms before implementation; record source version
Strong similarity to existing research	Narrow the contribution and compare the closest methods explicitly
Unknown initial temperatures	Separate cold-start results and expose wider uncertainty
Model exploits unavailable inputs	Enforce an input allowlist and test it automatically
Narrow bounds with missed hot events	Evaluate underestimation and coverage together; report failures
Long intervals make advice unusable	Report width and derating cost, not coverage alone
Optional simulator dominates schedule	Freeze the estimation scope first; keep control as an extension

Definition of a completed core project

The project is complete when another student can acquire the permitted data, reproduce the split and baseline results, replay a held-out fault scenario, and regenerate the main tables and figures. The report must include clean performance, F1-F3 robustness, unreported-error stress tests, uncertainty behavior, and representative worst cases.

The proposed 15% underestimation-reduction and 10% clean-error-tolerance targets guide interpretation; failing a target does not justify hiding results. Inference timing must name the tested hardware and batch size. Keep automotive certification, fleet deployment, measured energy savings, and prevented motor damage out of the claims unless independently supported.

First implementation session

Acquire the dataset, record its license and checksum, inspect profile lengths and temperature ranges, freeze the input allowlist, create the grouped split, and run the simple dynamic baseline. Review those results before selecting neural-model capacity or thresholds. This sequence makes the next decision evidence-based.

5. Research paper and presentation plan

Proposed research-paper structure

1. **Abstract:** problem, proposed method, dataset, and measured outcomes after experiments.
2. **Introduction:** thermal monitoring problem, sensor availability, and precise contribution.
3. **Related work:** recurrent estimators, thermal neural networks, hybrid models, and robustness studies.
4. **System and data:** input availability, target definition, provenance, grouped split, and initial conditions.
5. **Method:** thermal dynamics, missing-input representation, training loss, and calibration.
6. **Experimental design:** baselines, fault generation, metrics, seed policy, and paired comparisons.
7. **Results:** clean performance, dropout robustness, bias/stuck stress, intervals, latency, and ablations.
8. **Discussion:** observability, uncertainty failures, same-motor limits, and optional control simulation.
9. **Conclusion:** supported findings and specific remaining work.
10. **References and reproducibility statement:** source versions, code availability, and data terms.

Do not fill the results section with estimated improvements. Until experiments run, label all result tables as planned schemas without numerical findings.

Planned figures and tables

- Motor component schematic and estimator architecture.
- Per-profile data-split summary and an input-availability table.
- Representative temperature traces with dropout intervals and upper bounds.
- Underestimation versus dropout duration, with per-profile variation.
- Coverage-versus-width and recovery plots, including failure examples.
- Model comparison and ablation tables.
- A control-performance table only if the separate simulator is completed.

Suggested 12-slide presentation

Use one slide each for: the electrical problem; the research question; existing work and proposed contribution; dataset and signals; thermal equation and model; failure scenarios; evaluation design; actual baseline results; robustness and uncertainty findings; the replay demonstration; limitations and optional control study; conclusions and references. Before experiments, use a proposal deck and replace result slides with planned analyses.

Evidence ledger and handover

Tag every claim as a source-backed fact, a proposed design choice, a measured project result, or a simulation result. Preserve those distinctions in the paper, slide captions, and dashboard. The final handover should include environment instructions, data-acquisition steps, configuration files, a complete example run, model checkpoints where redistribution is permitted, and a short demo script.

Immediate next milestone

The first review should contain the data audit, exact profile split, baseline error plots, and the final frozen fault protocol. Those artifacts will determine whether the physics-guided model and uncertainty layer add enough value for the remaining implementation effort.

References and evidence status

Checked on 24 October 2026. These sources support dataset provenance and existing methods. The proposed robustness design, schedule, and success criteria are project choices requiring experiments.

R1. Primary dataset

Wilhelm Kirchgassner, Oliver Wallscheid, and Joachim Bocker. *Electric Motor Temperature*, dataset version 2161054, 2021. DOI: 10.34740/KAGGLE/DSV/2161054.

https://www.kaggle.com/dsv/2161054

The public card describes approximately 185 hours of 2 Hz PMSM test-bench measurements and profile identifiers. The data archive, exact schema, and license must still be audited at acquisition. A public dataset page does not by itself authorize unrestricted redistribution.

R2. Deep motor-temperature baseline

W. Kirchgassner, O. Wallscheid, and J. Bocker. *Estimating Electric Motor Temperatures With Deep Residual Machine Learning*. IEEE Transactions on Power Electronics, 36(7), 7480-7488, 2021. DOI: 10.1109/TPEL.2020.3045596.

https://doi.org/10.1109/TPEL.2020.3045596

Authors' implementation: https://github.com/upb-lea/deep-pmsm

The authors' repository identifies the journal issue year as 2021. The DOI contains 2020, reflecting the earlier publication history. This proposal follows the authors' issue citation.

R3. Thermal neural network foundation

W. Kirchgassner, O. Wallscheid, and J. Bocker. *Thermal neural networks: Lumped-parameter thermal modeling with state-space machine learning*. Engineering Applications of Artificial Intelligence, 117, 105537, 2023. DOI: 10.1016/j.engappai.2022.105537.

https://doi.org/10.1016/j.engappai.2022.105537

Authors' implementation: https://github.com/wkirgsn/thermal-nn

Preprint: https://arxiv.org/abs/2103.16323

This establishes the heat-transfer-based neural modeling foundation. The project must cite it and document any implementation reuse and license requirements.

R4. Recent hybrid-model prior art

OLTEM: Lumped Thermal and Deep Neural Model for PMSM Temperature. AI, 6(8), 173, 2025. DOI: 10.3390/ai6080173.

https://www.mdpi.com/2673-2688/6/8/173

This is a nearby hybrid-model study, so combining a thermal model with a neural network is not sufficient as a novelty claim. Verify full bibliographic metadata when building the final manuscript bibliography.

R5. Recent sparse-sampling prior art

A Physics-Informed Neural Network for PMSM Temperature Estimation Under Sparse Sampling Conditions. Energies, 19(15), 3701, 2026. DOI: 10.3390/en19153701.

https://www.mdpi.com/1996-1073/19/15/3701

The title and publisher record establish nearby 2026 work. Sparse sampling and loss of an entire input channel are different experimental problems. Read the complete method before deciding which claims or comparisons the final paper requires.